

Network slicing in private 5G for differentiated QoS in manufacturing

Manufacturing × Network automation and assurance × Network slicing · OS080 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
75 is the world moving	72 can we play, can we win	MEDIUM 1 named in the evidence	€2.9bn partial evidence · per year	LATER research and standards activi...	L0 16 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

Network slicing in private 5G lets manufacturers run different applications—from robotic control to video analytics—on one physical network, each with its own guaranteed quality of service. This opportunity is for manufacturing CIOs and operations leaders who need deterministic connectivity for Industry 4.0 use cases. It matters now because 5G slicing has moved from research to practical pilots, and the first standards and optimization techniques are mature enough to deploy.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€4.0bn €504m – €26.0bn	€2.9bn €362m – €18.6bn	€171m €21.7m – €1.1bn	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€1.3m €1.3m – €3.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Network slicing	30.0 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for Network slicing: No slicing series; industrial IoT use as proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

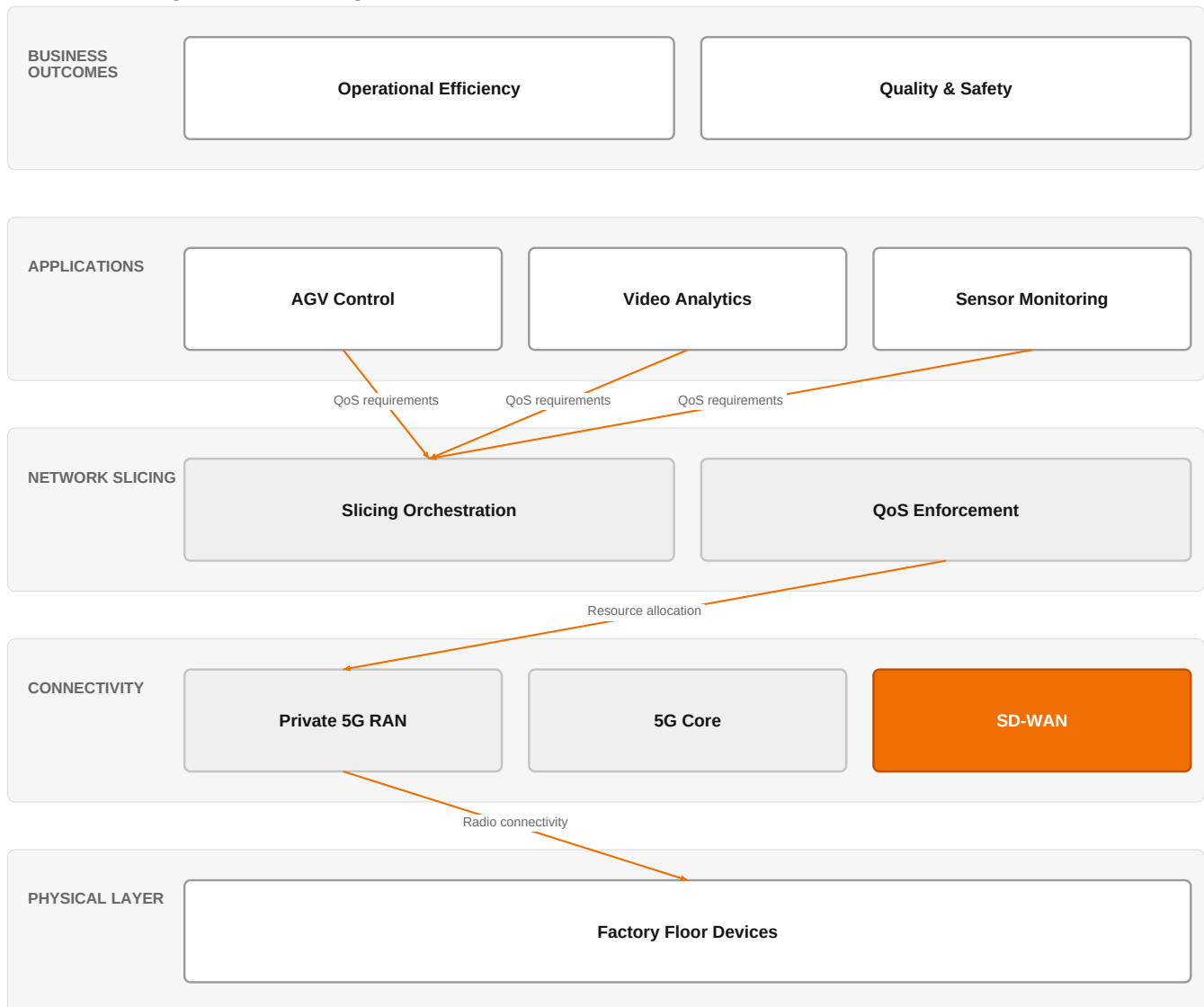
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Private 5G Slicing for Manufacturing



■ Orange asset
 ■ Partner
 ■ Customer-owned
 ■ Third party / to be sourced

This diagram shows how network slicing in a private 5G network enables different applications to receive tailored QoS, supported by Orange's orchestration and SD-WAN.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Private 5G networks are shifting from best-effort to deterministic, service-oriented architectures, with network slicing enabled by SDN and NFV allowing logical division of a single physical infrastructure. Research is actively addressing dynamic resource allocation and strict latency requirements through deep reinforcement learning, and 6G technologies are being piloted for resilient industrial manufacturing environments. AI-driven slicing is being explored for disaster-aware 6G infrastructures, indicating a trajectory toward more intelligent and resilient slicing.

Sources: SIG-DDBBDA6A8C40 International Scientific 2026-06-19; SIG-01FAF47D27BD IAR Journal of Engineer 2026-03-22; SIG-072593234114 cordis.europa.eu 2026-06-01; SIG-EBB2157E5B89 Acta Scientific Medica 2025-11-01

Who buys, and why now

The manufacturing operations director or plant IT manager feels the pain of managing multiple connectivity needs—AGVs, robots, sensors, and staff devices—each with different latency and reliability requirements, often over separate networks. The CIO signs the deal, triggered by a new automation project or a quality incident caused by network congestion. They need to guarantee that critical control traffic is never delayed by less critical data, without building multiple physical networks.

Why it is hot now — every claim bound to a dated source

- Network slicing within private 5G enables enterprises to offer different quality of service levels for various applications and use cases. [SIG-DDBBDA6A8C40]
- Intelligent network slicing optimization using deep reinforcement learning addresses dynamic resource allocation and strict latency requirements. [SIG-01FAF47D27BD]
- 6G technologies are being piloted for resilient industrial manufacturing environments. [SIG-072593234114]
- AI-driven network slicing for disaster-aware 6G infrastructures enhances resilience. [SIG-EBB2157E5B89]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a private 5G solution using its Evolution Platform as the orchestration layer, combined with Flexible SD-WAN for wide-area integration. IoT and Digital experts would design the slicing architecture, while Defense and security experts ensure compliance with ISO 27001 and TISAX certifications. Orange Group researchers would bring the latest optimization techniques. The engagement would start with a workshop to map application QoS needs, then design and deploy the sliced network, and finally operate it with continuous optimization.

Why Orange can win

Orange can win because it combines network expertise with deep industry knowledge, as evidenced by references like Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass. Its certifications (ISO 27001, TISAX) address the security and compliance concerns of manufacturers. However, the asset list is thin on specific private 5G or slicing offers, so Orange would need to leverage its research and integration capabilities to differentiate.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Evolution Platform	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
Flexible SD-WAN	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 7.4; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include network equipment vendors like Ericsson, Nokia, Huawei, and Boldyn Networks, who sell slicing as part of their infrastructure. Telecom operator peers like Colt, Deutsche Telekom/T-Systems, and Vodafone Business also offer slicing and are active in manufacturing. Accenture is active in manufacturing as a systems integrator. Ericsson is

specifically noted for highlighting monetisation opportunities for telcos. The customer will compare Orange's ability to integrate and manage the end-to-end solution against these players.

Sources: SIG-345DA62723A7 ET Telecom 2026-06-23

Competitor	Category	Position here	Basis	Evidence
Ericsson	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions	evidenced	SIG-345DA62723A7
Colt Technology Services	Telecom operator peer	sells Network slicing; competes in Connectivity Solutions	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Network slicing; active in Manufacturing; competes in Connectivity Solutions	structural	—
Vodafone Business	Telecom operator peer	sells Network slicing; active in Manufacturing; competes in Connectivity Solutions	structural	—
Boldyn Networks	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions	structural	—
Huawei	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Network slicing; active in Manufacturing; competes in Connectivity Solutions	structural	—
Accenture	Global systems integrator	active in Manufacturing	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What are the top three applications on your factory floor that require different levels of latency or reliability, and how are they currently connected?
- 2 Have you experienced any production incidents caused by network congestion or interference? If so, what was the impact?
- 3 Are you planning to deploy new automation or robotics that will increase the number of connected devices?
- 4 How do you currently manage QoS for your critical vs. non-critical traffic? Is it manual or automated?
- 5 What are your security and compliance requirements for your industrial network (e.g., ISO 27001, TISAX)?
- 6 Who in your organization is responsible for network infrastructure decisions, and what is their biggest pain point?

Objections you will meet

They will say	You can answer
We already have a reliable Wi-Fi network; why do we need private 5G and slicing?	Wi-Fi is best-effort and can suffer from interference and congestion. Private 5G with slicing provides deterministic QoS, which is essential for critical applications like AGVs or robotic control. It also allows you to consolidate multiple networks into one.
Network slicing sounds complex and unproven in manufacturing.	It's true that slicing is still evolving, but there are already pilots and research demonstrating its feasibility. Orange can bring its research expertise and integration experience to de-risk the deployment, starting with a small pilot to prove value.
We are concerned about the cost of deploying a private 5G network.	While there is an upfront investment, slicing can reduce long-term costs by eliminating the need for multiple separate networks. It also enables new use cases that can improve efficiency and reduce downtime, providing a strong ROI.

Risks and unknowns

The main risk is that network slicing in private 5G is still in early adoption, with standards and optimization techniques evolving. There is uncertainty about the actual demand from manufacturers and the willingness to pay for differentiated QoS. Additionally, the asset list lacks a dedicated private 5G offer, so Orange may need to partner or build capabilities, which could delay time-to-market.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing prospect, incorporating network slicing capabilities and highlighting our ISO 27001 and TISAX certifications to address security and compliance concerns, and referencing Saint-Gobain Glass as a relevant deployment.
Sales	In a conversation with a manufacturing client, mention how network slicing in private 5G can enable differentiated quality of service for various applications, referencing our work with Colgate-Palmolive and Mondelez International to illustrate real-world applicability.
Strategist / Innovator	Commission a deep-dive brief on network slicing optimization using deep reinforcement learning for private 5G in manufacturing, leveraging Orange Group researchers and IoT experts, to assess feasibility and potential for differentiated QoS.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-D8CAA774FDC4	2026-07-01	Next Research	1	Securing 5G network slicing for the smart grid network
SIG-DDBBDA6A8C40	2026-06-19	International Scientific ...	1	Private 5G and Network Slicing for Differentiated Services: Exploring How Network Slicing Withi...
SIG-072593234114	2026-06-01	cordis.europa.eu	1	Pilot for Resilient Industrial Manufacturing Environments with 6G Technologies
SIG-01FAF47D27BD	2026-03-22	IAR Journal of Engineerin...	1	Intelligent Network Slicing Optimization in 5G/6G Networks Using Deep Reinforcement Learning
SIG-C1DEE33668FF	2026-03-02	2026 21st Wireless On-Dem...	1	Proactive Resource Optimization for Heterogeneous 5G Network Slicing
SIG-048860EE76A9	2026-01-01	IEEE Access	1	Evolution of Industry 5.0 using AI based AMRs in Industrial Private 5G Network
SIG-2373B3F43C61	2026-01-01	IEEE Wireless Communicati...	1	Quantisation-Aware RIS Placement Optimisation for AGV Connectivity in Industrial Private 5G Net...
SIG-88237402830B	2025-11-28	SN Computer Science	1	Optimizing 5G Network Slicing for Heterogeneous Services
SIG-EBB2157E5B89	2025-11-01	Acta Scientific Medical S...	1	Resilient AIDriven Network Slicing for DisasterAware 6G Infrastructures
SIG-77D15EBBA923	2025-10-27	openalex.org	1	QoS-Aware Proportional Fairness Scheduling for Multi-Flow 5G UEs: A Smart Factory Perspective
SIG-F35834644774	2025-09-01	openalex.org	1	Emulating Industrial 5G-Edge Networks
SIG-A311539B246C	2025-08-22	2025 2nd International Co...	1	An Intelligent Framework for Network Slicing to Enhancing Resource Scheduling in 5G Network
SIG-345DA62723A7	2026-06-23	ET Telecom	2	5G network slicing can create new monetisation opportunities for telcos: Ericsson's Nitin Bansa...
SIG-77F434506B29	2026-07-29	framerusercontent.c om	3	Industry Brief: Private 5G for Manufacturing and Industrial Sites [pdf]
SIG-67C4E4DA4852	2026-07-01	arxiv.org	3	Latency-Sensitive 5G RAN Slicing for Industry 4.0
SIG-AF0B9D624A57	2026-06-30	arxiv.org	3	Latency-Sensitive 5G RAN Slicing for Deterministic Aperiodic Traffic in Smart Manufacturing

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:43+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.